

The State-Dependent Phillips Curve:

Evidence from Machine Learning

Ece Taşan, Emilia Infante, Orhun Özel, Marta Bratkowska, Zeynep Pehlivan

Barcelona School of Economics — Master's Thesis

Supervisors: Joan Monràs & Rodrigo Carril

June 2026

Overview

- ① Motivation
- ② Literature
- ③ Methodology
- ④ Results
- ⑤ Conclusion

One Linear Phillips Curve Does Not Fit All

- ① **Non-Shock Periods in 1990s-2020:** inflation–unemployment link was weak
- ② **GFC:** deep recession, models predict deflation → **never comes**
- ③ **COVID:** tight labour markets, models predict modest inflation → **hits nearly 9%**

Is this caused by nonlinearities and state dependence?

Key insight

Anchored expectations masked the true slope.

Under large shocks the relationship **switches back on**.

Why it matters

- Inflation forecasts drive **policy rates** and forward guidance
- Systematic errors erode **credibility** and delay response

What the Literature Says

What happened to the Phillips Curve?

Camp 1: Stable Slope

Flatness reflects anchored expectations, not a structural change (Jørgensen & Lansing, 2025).

Post-pandemic inflation was driven by goods shocks; no PC nonlinearity (Bernanke & Blanchard, 2023).

Camp 2: Non-Linear Form

L-shaped PC: Flat when $v/u < 1$, steep above (Benigno & Eggertsson, 2023).

Nonlinear terms (u^2 , u^3) capture convexity (Ball, Leigh & Mishra, 2022).

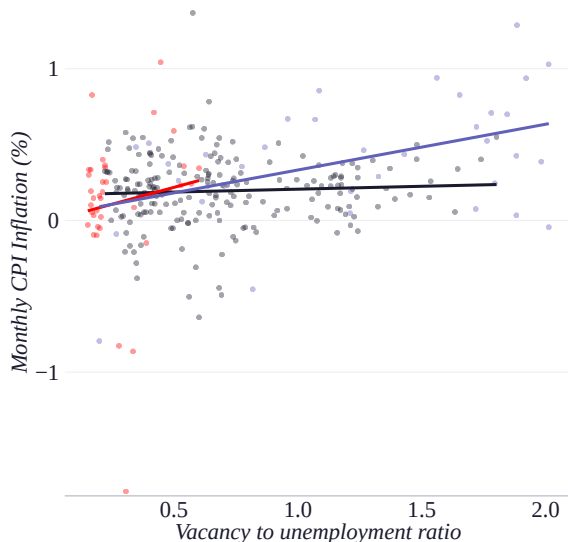
Camp 3: State-Dependent

Labour tightness alone is not enough.

The slope shifts with supply stress, slack, business cycle, and shock type. The dominant driver changes by episode. (Demirel & Wilson, 2023; Coulombe, 2024)

Descriptive Evidence: The Slope Switches On

- Normal Periods
- 2008–10 GFC
- 2020–22 COVID



The dominant driver rotates:

- **Calm years:** Essentially flat, slack barely moves inflation
- **GFC:** Slope is steeper, yet deflation never comes (may be non-linear, $v/u < 1$)
- **COVID:** Steep positive slope, tight labor market ($v/u > 1$)

A single fixed slope, estimated mostly on calm years, cannot predict well when the regime changes.

Research Question, Hypothesis & Contribution

Research Question

Do more flexible ML models (non-linear and/or state dependent) outperform linear PC models *specifically during shock episodes*? How much of the gain is due to nonlinearity vs. variable selection?

Hypothesis

Linear models fail during shocks because they cannot adapt to regime changes. ML detects when the slope changes and which drivers dominate $\Rightarrow$ forecast gains $\equiv$ **indication of state-dependency**.

Contribution

ML as an **agnostic diagnostic** (no functional form assumed); a 2×2 design that **disentangles** nonlinearity from variable selection; a variable-importance analysis showing the **active drivers rotate** across states.

The 2×2 Model Design

	Labor Market Indicators Only	Full FRED-MD (127 series)
Linear	Linear Phillips Curve LASSO/Ridge/Elastic Net Baseline	Linear + Variable Selection LASSO/Ridge/Elastic Net Includes omitted variables
Non-linear	Non-linear Phillips Curve Random Forest & LLF Flexible functional form	Non-linear + Var. Selection Random Forest & LLF Both margins combined

left → **right**: isolates selection
top → **bottom**: isolates nonlinearity

Data

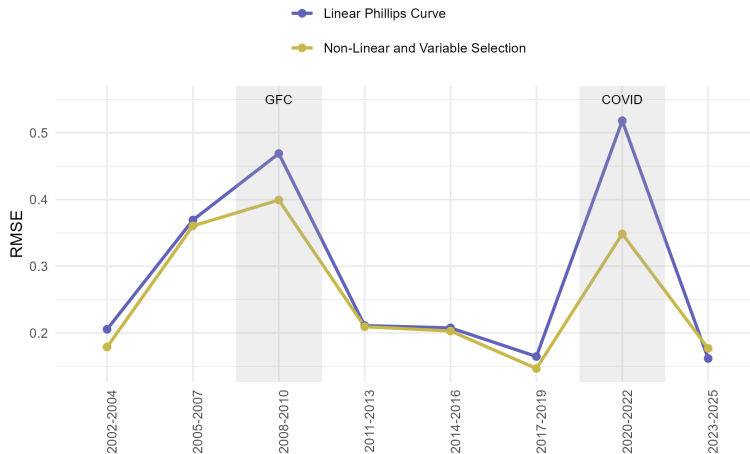
- FRED-MD: 127 monthly series (1960–2025)
- Target: monthly CPI inflation; eval. 2001–2025
- Shocks: 2008–10 GFC, 2020–22 COVID
- Rolling-window out-of-sample forecasts

What we compare

- Root Mean Square Errors (RMSE)
- Is the gap larger *during* shock episodes?
- Decompose: nonlinearity vs. variable selection
- Variable importance: what does ML upweight in shocks?

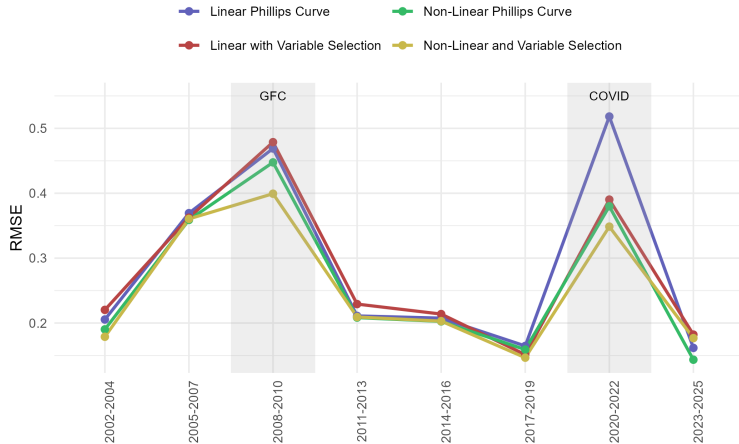
RMSE gain during shocks $\equiv$ signal of state-dependence

Main Comparison: Forecast Performance Across Regimes



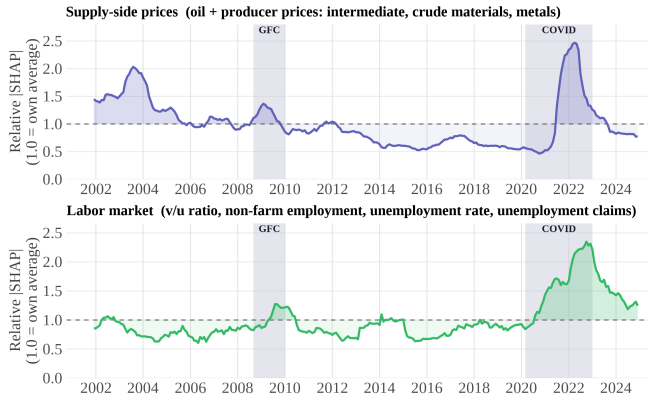
Models perform similarly in calm periods, the gap opens sharply in large shocks.
The gains from Non-lin and Var.Sel. model in GFC and COVID is 15% and 33% respectively.

Main Comparison: Forecast Performance Across Regimes



Both channels are important. Var.Sel. becomes important in COVID period, in line with supply side indicators. Non-linearity is especially important in GFC. The combination of both channels gives the best results.

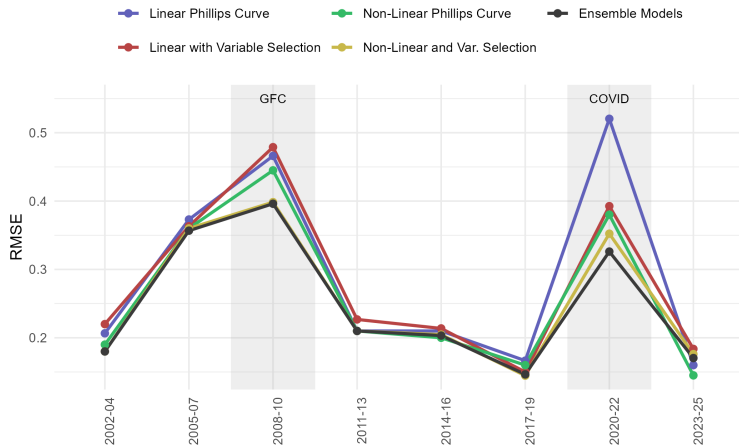
Variable Importance: Drivers Rotate Across Regimes



SHAP plot indications

- Supply-price and labour-market groups more than double their contribution in 2021–22
- Wages: Wages move freely up in tight labor markets (capacity). Wage rigidities in slack markets. (Benigno et al., 2023)
- Inputs: Large disruptions amplify the effect of inputs, create dependency on const. input (Baqae and Farhi, 2022)

Ensemble Forecasting



The further gains from ensemble models is around 2% on average.

Comparison with Cleveland FED Forecast

Out of Sample RMSE

1-Months Ahead – Ensemble vs. Cleveland FED

Year	Ensemble Models	Cleveland FED
2011-2013	0.121	0.206
2014-2016	0.173	0.190
2017-2019	0.143	0.132
2020-2022	0.316	0.358
2023-2025	0.164	0.121
Average All	0.183	0.201

Comparisons

- **On Average:** Ensemble models lowers RMSE by $\approx 8.9\%$
- **COVID 2020–22:** Ensemble models lowers RMSE by $\approx 11.7\%$

Main Conclusions

Both the **predictors** and their **mapping into inflation** change with the shock.
Single PC slope cannot capture this.

What we find

- **PC is state-dependent:** Flexible models outperform the linear PC specifically in the GFC and COVID windows (-15% and -33% RMSE); in calm years all models perform similarly.
- **Both margins matter:** both non-linearity and variable selection contribute to the prediction gains.
- **Drivers rotate:** supply prices and labour tightness contributed more heavily to the prediction during COVID.

Why it matters

- The results contribute to the debate on the shape of the PC **without prior assumptions**.
- Inflation forecasts drive **policy rates** and forward guidance.
- Forecasting needs **regime-awareness**: a fixed slope estimated on calm years misleads when supply chains break or labour markets overheat.

References

- Ball, L., Leigh, D., & Mishra, P. (2022). *Understanding U.S. inflation during the COVID era*. Brookings Papers on Economic Activity.
- Benigno, P., & Eggertsson, G. (2023). *It's baaack: ... the return of the non-linear Phillips curve*. NBER WP 31197.
- Bernanke, B., & Blanchard, O. (2023). *What caused the U.S. pandemic-era inflation?* Hutchins Center WP 86.
- Coulombe, P.G. (2024). *The macroeconomy as a random forest*. Journal of Applied Econometrics, 39(3).
- Demirel, U.D., & Wilson, M. (2023). *Effects of fiscal policy on inflation: supply disruptions and economic slack*. CBO WP 2023-05.
- Gitti, G. (2024). *A nonlinear regional Phillips curve*. Working Paper.
- Harding, M., Lindé, J., & Trabandt, M. (2023). *Resolving the missing deflation puzzle*. Journal of Monetary Economics.
- Jørgensen, P.L., & Lansing, K.J. (2025). *Anchored inflation expectations and the slope of the Phillips curve*. European Economic Review.
- Lucas, R.E. (1976). *Econometric policy evaluation: A critique*. Carnegie-Rochester Conf. Series.
- McCracken, M.W., & Ng, S. (2016). *FRED-MD: A monthly database for macroeconomic research*. JBES, 34(4).
- Medeiros, M.C., Vasconcelos, G., Veiga, Á., & Zilberman, E. (2021). *Forecasting inflation in a data-rich environment: the benefits of machine learning*. JBES, 39(1).
- Michailat, P., & Saez, E. (2024). $u^* = \sqrt{uv}$: *The full-employment rate of unemployment*. Brookings Papers on Economic Activity.

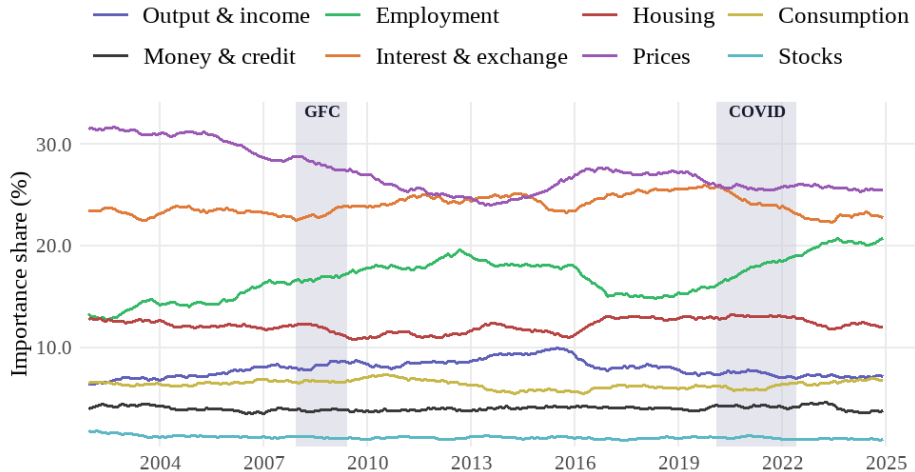
Thank you

Comments and suggestions welcome.

Ece Taşan, Emilia Infante, Orhun Özel, Marta Bratkowska, Zeynep Pehlivan

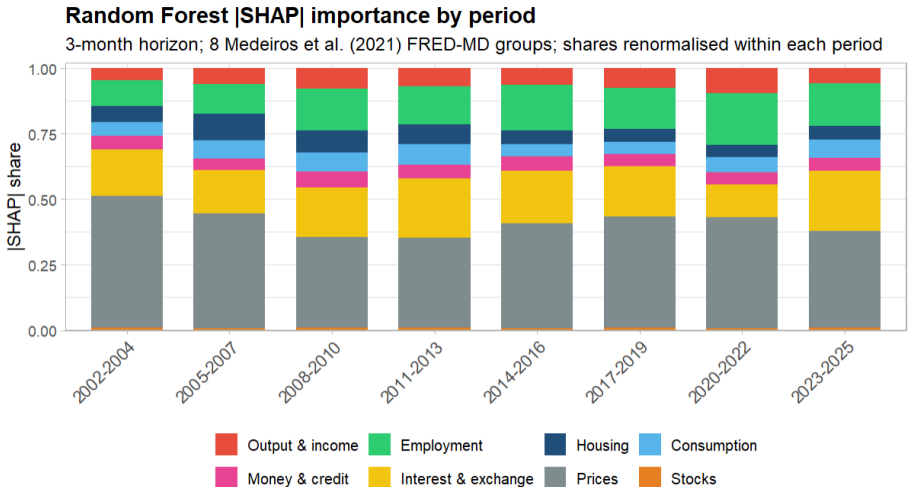
· BSE Master's Thesis · June 2026

Appendix 1: Variable Importance (12-month rolling window)



The further gains from ensemble models is around 2% on average.

Appendix 2: SHAP Importance by Period



The further gains from ensemble models is around 2% on average.